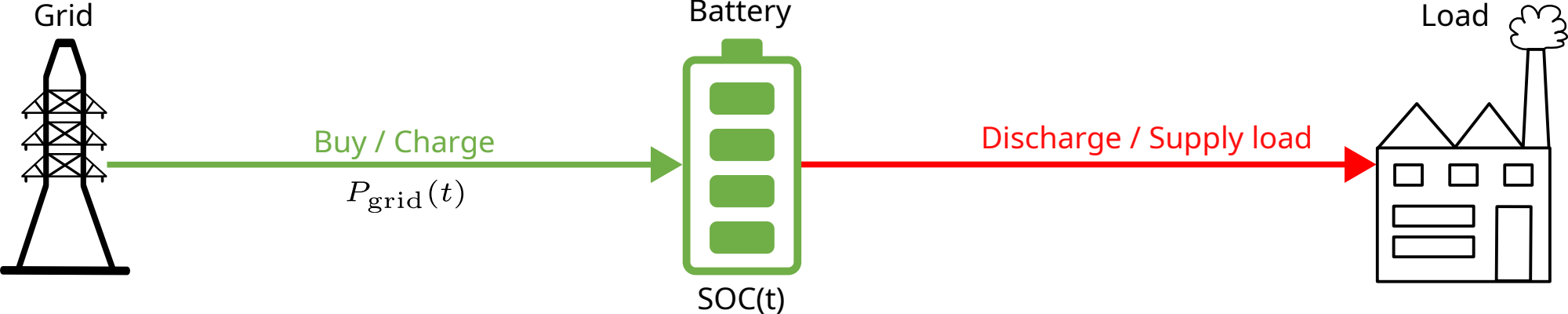
Grid
Battery
Load
Buy / Charge
$P_{\mathrm{grid}}(t)$
Discharge / Supply load
SOC(t)